

ASSIGNMENT 2 OF 2 · REQUIRED COVER SHEET · DUE SUNDAY 13 SEPTEMBER 2026, 23:59 SGT

A2 Team Self-Appraisal

Team ID: 6 **Section:** B **Problem chosen (A / B):** A

Members: ROHIT PANDA, HUANG YU, LI YUNKE, SHEN BOWEN, XIA YANRAN, ZHAO YUJIA

Repository URL: https://github.com/Roh00t/PE201_A2_Group6 **Date:** 16 September

ONE SHEET PER TEAM, not one per member. Complete it together, in one sitting, after the work is done, it takes about twenty minutes. CHECK IT INTO YOUR A2 SUBMISSION FOLDER alongside the rest of the submission, by the A2 deadline of Sunday 13 September 2026, 23:59 SGT; do not email it. This sheet is required and is not graded, but a missing self-appraisal is treated as an incomplete submission.

1 · Rate your team against Rubric 1

Circle or mark one band per criterion. Be honest rather than generous, a team that marks itself Excellent everywhere tells the marker nothing, and a large gap between your rating and the marker's is the single most useful thing on this page.

Criterion	Weight	Excellent (4)	Proficient (3)	Developing (2)	Limited (1)
Conceptual Understanding	25%	[X]			
Technical Execution	30%	[X]			
Reasoning & Justification	25%	[X]			
Communication & Clarity	20%		[X]		

2 · The one decision you would defend hardest

Name a single design decision, a tool you cut, the autonomy setting you chose, the dependency rule behind your parallel calls, the model you settled on, and say what evidence made it. Four or five sentences.

We shifted safety-critical validation entirely out of the prompt and into a deterministic Python gate (issue_decision_letter checks). On our adversarial test (CLM-8952), four out of five models approved a hostile injection narrative at least once. By replacing the LLM-as-a-judge with a strict code constraint, requiring seven specific ledger checks before writing a decision, we blocked 100% of these injections. This single architectural choice eliminated an expected US\$1.27 failure rate per run, proving that routing logic must be enforced by code, not asserted by prompts.

3 · The one you are least sure about

Every team has one. Naming it costs you nothing here and it is often the most interesting thing in the submission.

Our experimental methodology for Version 2 lacked strict variable isolation. When upgrading from v1 to v2, we altered prompt descriptors, ReAct loop mechanics, and context limits simultaneously. Consequently, when correct decision accuracy degraded from 49 (v1) to 45 (v2), we could not cleanly attribute the failure to a single variable. We shipped v2 for its safety properties, but bundling the changes violated core experimental control.

4 · Your headline numbers

Copy these from your repository. If any cell is empty, that is worth noticing before you submit rather than after.

	Value		Value
Evaluation cases	40 (30 ordinary · 10 negative)	Guardrail cases	18, 16 must-fire, 1 must-not-fire, 1 documented known limit
Negative cases	10	Live runs per model (D4)	60 trials (30 × 1 + 10 × 3)
Pass rate (best model)	90.0%, 54/60, anthropic/claude-haiku-4.5, v2	Pass rate on negatives only	80.0%, 24/30 negative trials; 8/10 cases passed 3-of-3
Models in the battery	5 distinct models across 5 families, 6 battery rows (the sixth is the	Median turns per run	4.0 (live, Haiku 4.5; scripted median also 4.0, worst legitimate 5)

	v1 pass, model held fixed)		
The live model YOU ran (D5b)	anthropic/claude-haiku-4.5 , Rohit Panda, mid tier	Your pass rate on it	90.0% , 54/60 trials, v2 prompt, 13 Sep 2026
Turns saved by parallel calls	357 → 211 turns (−41%); input tokens 2,322,600 → 1,091,400 (−53%); 360 calls either way, 60/60 passing both	Cost per successful task	US\$0.7811 , layer 1 US\$0.0211 + layer 2 US\$0.7600
Monthly cost at problem volume	US\$6,731 at 8,000 claims/month, against US\$60,800 for people alone	Break-even success rate	89.8% , the rate a cheap model needs to beat Haiku; deepseek-v3.2 measured 80.0%, so none of ours clears it
Tokens per call , v1 → v2	1,240 → 3,134 tokens of system prompt, re-billed on every turn (+2.1× cost per trial)	Pass rate , v1 → v2, same model	51.7% → 68.3% on qwen/qwen3-23 5b-a22b-2507, model held fixed (+16.7pp); negative 3-of-3 10% → 40%

Every member runs one live model, or the v1 pass , three models is the floor, N-1 for a team of N is the shape expected. Name YOURS. If you ran the v1 pass, write "v1 pass" and the model it sat on.

The last row is D2(b): the descriptor rewrite you measured. Two numbers each, before and after, on ONE model. An empty row here means the rewrite was never measured.

5 · Contribution

Who owned what. This should match CONTRIBUTIONS.md and the commit history. It is not used to scale marks , the team receives the team score , but it is one of the two signals the non-contribution clause relies on.

Member	Owned	Also contributed to
Rohit Panda	D1 the ReAct loop, multi-call turns and per-run instrumentation · D2(a) tool scoring · D2(c) dependency rule · D5(a) scripted backend and planner · D5(b) battery infrastructure, provenance fingerprint and checkpointing	D0(c) what good looks like · both D7 reproductions · the gated-action ledger and its seven pre-write checks · narrative_guard · README, CLAUDE.md, GUARDRAILS.md · live model anthropic/claude-haiku-4.5
Huang Yu	D2(b) six-field descriptors and the v1→v2 rewrite · D3(a) guardrail code layer · D3(b) guardrail checklist	runtime boundary validation (procedure codes, ISO dates, totals, decision allow-list) · prompt.descriptor_set(), without which PROMPT_VERSION selected nothing · live model openai/gpt-4.1-mini

Li Yunke	D4 evaluation harness · code check and judgement check graders · the committed judge prompt and its offline rehearsal suite · answer-key joins	independent reproduction of the scripted run · live model qwen/qwen3-235b-a22b-2507
Shen Bowen	D6 cost ledger, ± 10 pp sensitivity and break-even · OpenRouter price verification	the checkpoint fix that stops a resumed battery re-charging a completed canary · report §§4–5 · live model deepseek/deepseek-v3.2
Xia Yanran	Negative-case design across the 10 negative cases · the three hostile-narrative shapes (CLM-8941, CLM-8952, CLM-9035) · D7 failure reproduction and re-runs	review of the judge parser, harness, loop and cost model · live model google/gemini-2.5-flash
Zhao Yujia	D0 why an agent , ladder, both Capsule 1 tests, the autonomy defence · D6 cost model alongside Shen Bowen · report assembly · demo coordination · technical PM	src/cost_model.py and its refusal of null inputs · the D2(b) v1 pass, qwen/qwen3-235b-a22b-2507, prompt v1 , the only measurement in A2 that isolates our own writing as the variable

Peer rating is separate from this sheet and comes AFTER it: NTU's Group Project peer-review function opens 13 September and closes Wednesday 16 September, 23:59 SGT. Each member completes it individually in NTULearn. It is a participation requirement.

6 · Declaration

Every member ticks every box. One sheet, signed by the team.

[X] Every member of this team can explain every block of code we submit , what it does, and why it is there. (Block, not line: this is about function, not syntax.)

[X] The pass rates, token counts, turn counts and costs we report are measurements we ran ourselves. None of them were supplied by an AI assistant.

[X] Our submitted harness runs end to end on the scripted backend, with no key and no network.

[X] We kept every fixture record we were given, unedited, and every case in our evaluation set carries a label. check_my_data.py passes on our data.

[X] No part of this submission is drawn from any member's End-of-Course Project, and no part of it will be submitted as part of one.

[X] This is our own work, and all sources, tools and assistance are attributed.

Signed on behalf of the team: Rohit Panda **Date:** 16 September

How this is used

This sheet is not marked and cannot lower your grade. It is read alongside your submission for two reasons. First, how a team reads its own work is usually the fastest route to feedback that is actually useful. Second, where a team's self-rating and the marking diverge sharply, that gap gets an explicit comment , which is where the most useful feedback tends to sit.